CMS TEAM PROJECT · TU DRESDEN · SOSE 2026

# Moderating the emotional tone of AI data stories

What we measured, on 32 runs across eight datasets, judged blind by Claude Opus 5 against 25 stories our own team wrote from the same data.

Experiments only · every figure generated from the run database

THE RESULT

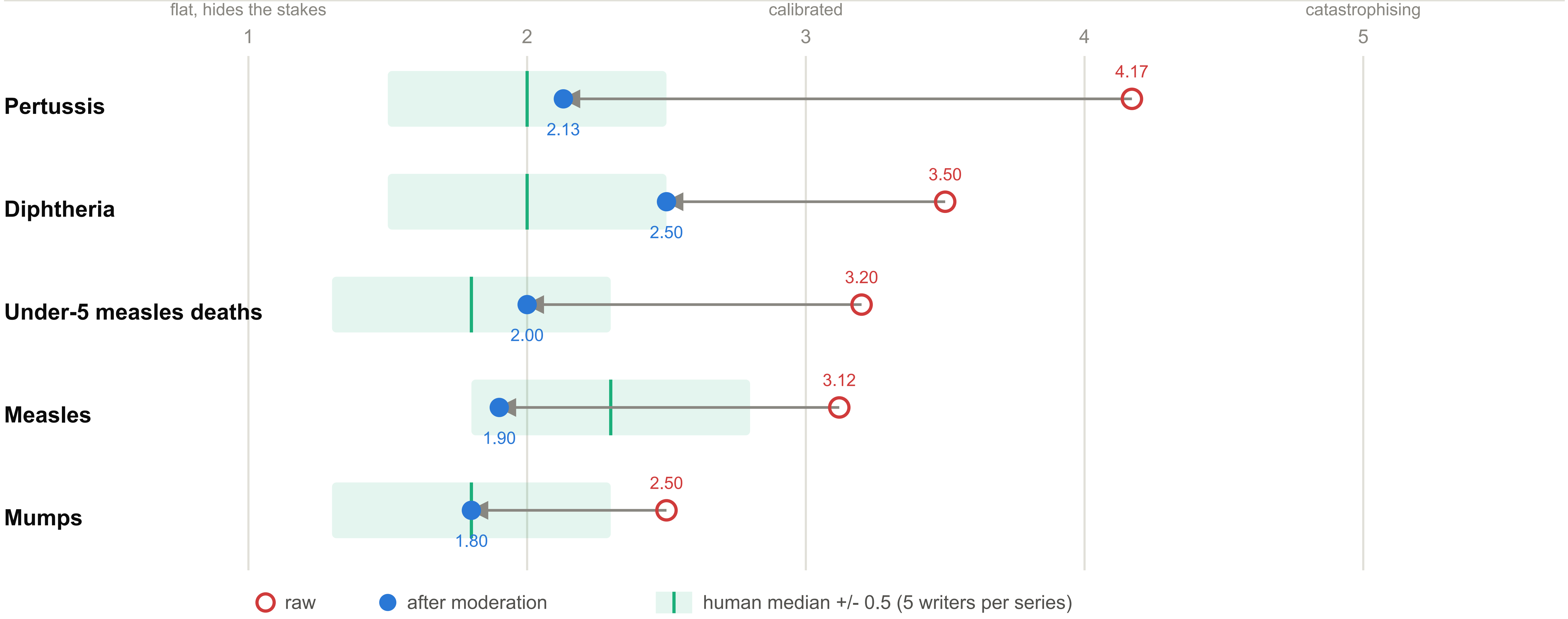
# Moderation lands machine stories on the human level

| RAW, ALARMISM | AFTER MODERATION | OUR WRITERS |
| --- | --- | --- |
| 3.74 | 2.09 | 2.00 |
| the generator's own tone | a 1.5-point move | 25 stories, same evidence packs |

Alarmism 1–5 · both ends are failures · 3 is calibrated

# Moderation moves machine stories onto the human level

Alarmism 1-5, Claude Opus judging blind, one story at a time. 32 runs against 25 human stories. Overall 3.74 -> 2.09, human median 2.00.




The runs are not spread evenly: pertussis carries 24 and measles carries 5, and the remaining series are a single run each, so those arrows are one story rather than a mean. The human stories carry no headline and the machine stories do; headlines concentrate alarmism, so the human line is if anything flattered and every gap shown is a lower bound.

WHY OUR METRIC HAS TWO AXES

# A falling series fails in the opposite direction

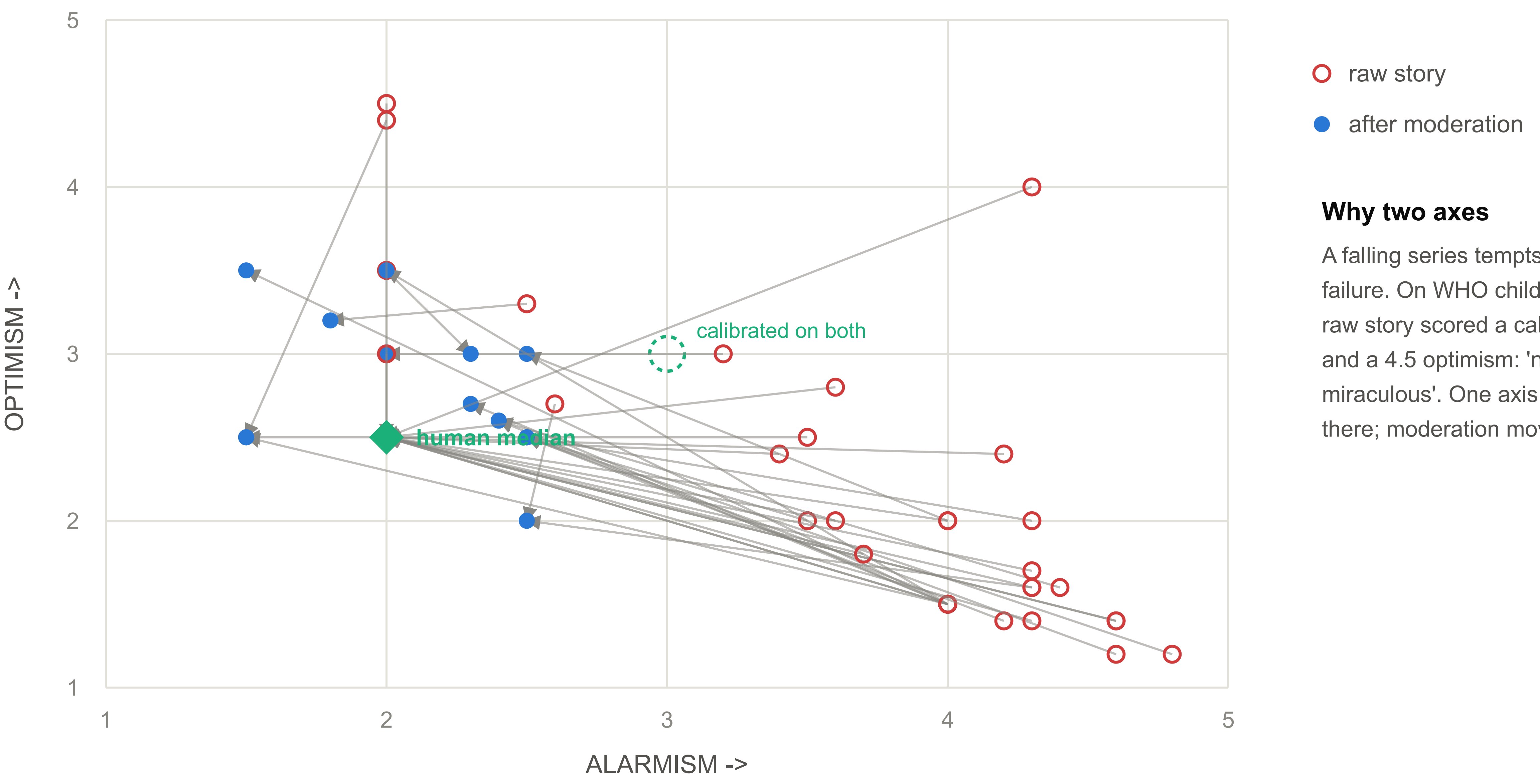
On WHO child mortality the raw story scored a calm **2.0** for alarmism while calling two decades of progress "nothing short of miraculous". Optimism caught it at **4.5**, and moderation moved it to 2.5.

A single-axis judge reports that moderation did nothing on that dataset.

alarmism 3.74 → 2.09 · optimism 2.15 → 2.60

# Two axes, because a falling series fails in the other direction

Every run: 35 raw-to-moderated moves, all eight datasets.




### Why two axes

A falling series tempts the opposite failure. On WHO child mortality the raw story scored a calm 2.0 alarmism and a 4.5 optimism: 'nothing short of miraculous'. One axis sees nothing there; moderation moved it to 2.5.

Alarmism falls 3.74 -> 2.09 and optimism rises 2.15 -> 2.60, both toward the middle. Note the moderated cloud settles below alarmism 3: the moderator aims at quiet rather than at calibrated, and lands near where these human writers also sit (diamond).

CHOOSING THE MODELS

# Six combinations tie on tone. Faithfulness breaks the tie.

The moderator converges on the same tone whoever wrote the draft (2.10 vs 2.10, $p = 1.00$). What separates the generators is how many of the raw story's figures survive.

QWEN3.5:4B × GEMMA4:31B

**61%**

figures kept, ± 20 over 5 seeds

LLAMA3.1:8B × GEMMA4:31B

**90%**

figures kept, ± 15 over 5 seeds

Welch $t = -2.7$ · $p = 0.0288$ · Cohen $d = 1.71$

THE LESSON WE NEARLY PUBLISHED

# With one run per cell, our ranking was backwards

A single run had qwen3.5:4b keeping **71%** of figures and llama3.1:8b keeping **50%**, so we recommended qwen3.5:4b.

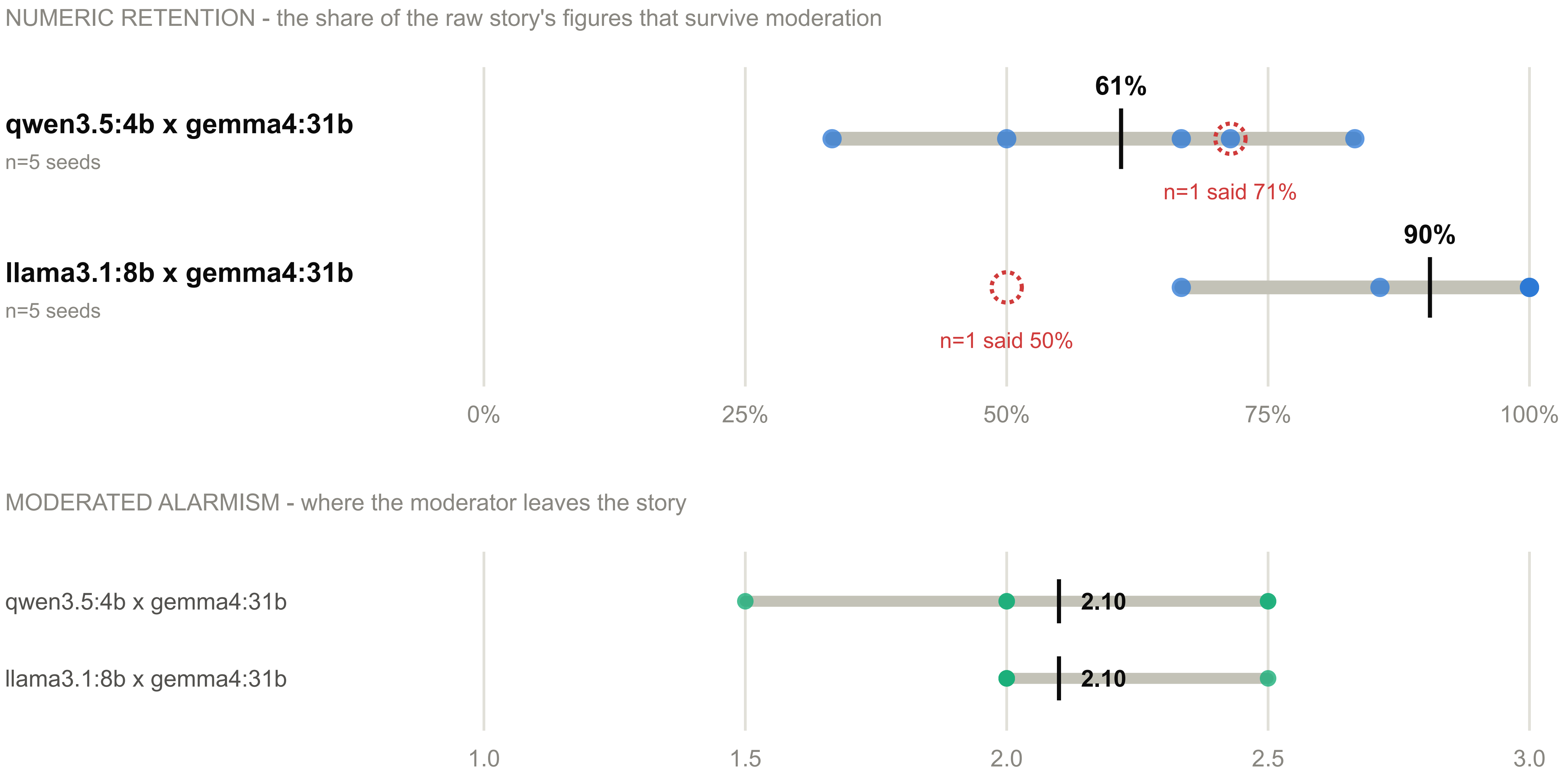
Five seeds each: **61%** and **91%**. Both original numbers were unrepresentative draws from distributions spanning 33-83% and 67-100%.

`n = 1 cannot tell a model from a draft`

Judge noise is 0.08 points · run-to-run spread is 0.42 · the variance is the generator's

# Five seeds per cell, and the recommendation flips

Same two configurations, run five times each at distinct seeds, pertussis-global.




**The ranking was backwards**

One run each had put qwen3.5:4b ahead on retention, 71% to 50%. Five seeds each reverse it: 61% to 91%.

Welch t = -2.70, p = 0.029, Cohen d = 1.71.

Identical: 2.10 both, p = 1.00. The moderator converges on the same tone whoever wrote the draft.

Seeds differ deliberately: repeats at a fixed seed replay the same text and measure determinism, not stability. Judge noise is not the source of this spread - three independent passes over the same story agree to ICC 0.99, mean spread 0.08, against a run-to-run alarmism spread of 0.42. The variance is the generator's, not the judge's.

CAN THE INSTRUMENT BE TRUSTED

# The tone scale has an error bar

ICC(2,1)

**0.991**

absolute agreement, 3 passes

KRIPPENDORFF A

**0.991**

interval scale

MEAN DISAGREEMENT

**0.08**

points, out of a 1-5 scale

Same model and prompt each pass, so this is self-consistency: an upper bound on what a different judge would agree to.

192 calls · 41/64 identical on all three passes

SCALE OR LINEAGE

# Writing calmly is a family trait, not a size effect

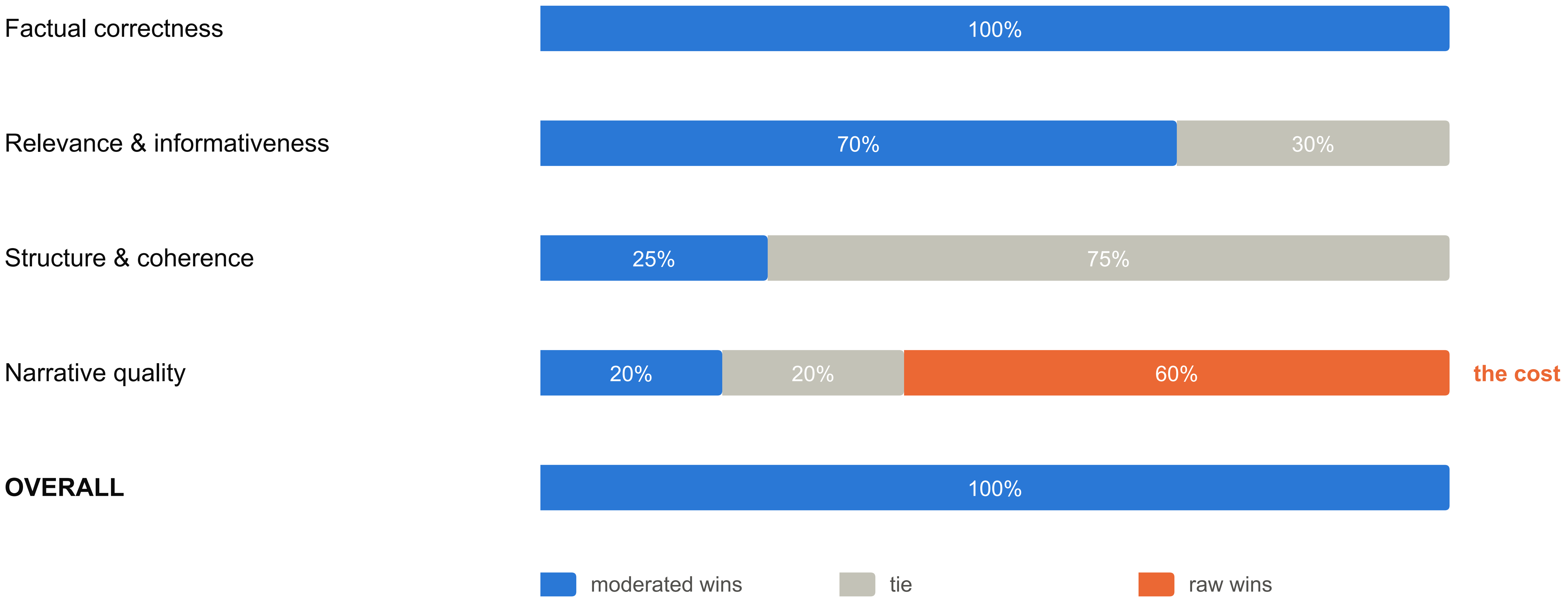
Raw alarmism across the qwen3.5 ladder: **4.8** at 2B, **4.6** at 4B, **4.5** at 9B, **4.3** at 27B. Scaling by more than 13× changes nothing.

The one calm generator in the set is **llama3.1:8b at 3.98**. Our earlier reading, that smaller generators write hotter copy, was a family effect misread as a scale one.

Same series, same seed, same moderator throughout

# The moderated story wins on everything except readability

20 blinded pairs, position randomised, Claude Opus deciding each criterion.




The judge is never told which story was moderated, and which one is shown first is decided by a coin flip per pair. Narrative quality is the one criterion the raw story wins, and it wins it three times as often: this system buys calibration and accuracy by spending engagement, and a deck that hides that row is selling rather than reporting.

WHAT IT COSTS, AND WHAT WE CANNOT CLAIM

# The honest slide

- **Narrative quality is the price.** The unmoderated story wins it three times as often. We buy calibration and accuracy by spending engagement.
- **n = 1 outside two cells.** Every other combination is a single run, and we have shown what that can do to a conclusion.
- **One judge.** We measured how much it agrees with itself, not how much a different judge would agree with it.
- **Seven datasets are about three.** Five are WHO series of the same shape and two are nested.

Full caveat list in the report · experiments/LOOPHOLES.md

IN ONE LINE

# Tone moderation reaches the human level, and the measurement holds up.

Alarmism 3.74 → 2.09 against a human 2.00, 100% on factual correctness, judge ICC 0.99. The costs are narrative quality and a sample size we are still honest about.